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## DESIGN: BASICS

that would work well with such high frequencies and for the entire range of frequencies (of various carrier waves) with same quality is very difficult.

Then how about having many amplifiers, each designed for various carrier frequencies? That may work, but it would make the cost and size of such a receiver very high. Remember, even such a solution demands all the amplifiers working for high frequencies as well. Imagine how difficult it could have been in those early days of AM broadcasting and reception where realising high frequency circuits was still a challenge. No wonder, the concept of IF emerged then.

### The concept of IF

So, what is the solution for the limitations described above? Well, how about creating an amplifier that is well suited for a single frequency and converting the received modulated carrier wave of any station to the frequency to which the amplifier is well suited for? It is like, let us say we have an amplifier designed to amplify a signal at 100kHz, and we convert the modulated wave received from 1200kHz to 100kHz so that the amplifier can happily amplify it. Similarly, if we need to hear the radio station transmitting the modulated carrier signal at 1400kHz, we simply convert the received modulated wave of 1400kHz to 100kHz and then send it to the amplifier that is designed to work well at 100kHz. So now we are converting any modulated carrier frequency of any station in the entire AM frequency range to a fixed 100kHz, and then such a signal at 100kHz is amplified, and then audio wave is extracted from that 100kHz. This frequency, that is, 100kHz in this example, is called the intermediate frequency or IF.

### How IF is achieved

Can it be done? It turns out that we can! How do we do that?

Let us see an even better example.

Imagine a radio transmitter or broadcasting station sending the modulated carrier wave at 300kHz. Let us also imagine that, at the receiver side, we have an oscillator, and let us call this oscillator as 'local oscillator' or simply LO (see Fig. 1). Assume the LO would produce any frequency we want. Remember, the LO's output need not be an amplified signal. Since the received modulated carrier signal, which we are going to mix it with, is weak anyway, and thus